MANCHESTER 1824

The University of Manchester

EART11200 - The Natural Scientist's Toolkit

Mathematics and Physics

Week 10 Practical - Vectors and Partial Differentiation (Solutions)

Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Describe the properties of scalar and vector fields and their applications in mathematics and physics

(2) Apply basic vector operations, such as addition, subtraction, scalar multiplication, and magnitude, in theoretical and real-world applications.

(3) Compute partial derivatives to determine rates of change of scalar fields in theoretical and real-world applications.

Exercise 1: Vector Calculations

- Suppose that $\vec{a}$, $\vec{b}$, and $\vec{c}$ are vector such that:
  - $\vec{a} = \langle 5, 3 \rangle$
  - $\vec{b} = \langle -2, 5 \rangle$
  - $\vec{c} = \langle -1, 3, 5 \rangle$
- Calculate the following:
  (a) $3\vec{c}$
  (b) $\vec{a} - \vec{b}$
  (c) $10\vec{b} - 3\vec{a}$
  (d) $|\vec{c}|$
  (e) $|\vec{b} - \vec{a}|$

Solution 1: Vector Calculations

- Suppose that $\vec{a}$, $\vec{b}$, and $\vec{c}$ are vector such that:
  - $\vec{a} = \langle 5, 3 \rangle$
  - $\vec{b} = \langle -2, 5 \rangle$
  - $\vec{c} = \langle -1, 3, 5 \rangle$
- Then we see that

(a)

$$\begin{aligned} 3\vec{c} &= \langle 3 \times (-1), 3 \times 3, 3 \times 5 \rangle \\ &= \langle -3, 9, 15 \rangle \end{aligned}$$

(b)

$$\begin{aligned} \vec{a} - \vec{b} &= \langle 5, 3 \rangle - \langle -2, 5 \rangle \\ &= \langle 5 + 2, 3 - 5 \rangle \\ &= \langle 7, -2 \rangle \end{aligned}$$

Solution 1: Vector Calculations

- Suppose that $\vec{a}$, $\vec{b}$, and $\vec{c}$ are vector such that:
  - $\vec{a} = \langle 5, 3 \rangle$
  - $\vec{b} = \langle -2, 5 \rangle$
  - $\vec{c} = \langle -1, 3, 5 \rangle$
- Then we see that

  (c)

$$\begin{aligned} 10\vec{b} &= \langle 10 \times (-2), 10 \times 5 \rangle = \langle -20, 50 \rangle \\ 3\vec{a} &= \langle 3 \times 5, 3 \times 3 \rangle = \langle 15, 9 \rangle \\ 10\vec{b} - 3\vec{a} &= \langle -20, 50 \rangle - \langle 15, 9 \rangle \\ &= \langle -20 - 15, 50 - 9 \rangle = \langle -35, 41 \rangle \end{aligned}$$

Solution 1: Vector Calculations

- Suppose that $\vec{a}$, $\vec{b}$, and $\vec{c}$ are vector such that:
  - $\vec{a} = \langle 5, 3 \rangle$
  - $\vec{b} = \langle -2, 5 \rangle$
  - $\vec{c} = \langle -1, 3, 5 \rangle$
- Then we see that

(d)

$$\begin{aligned} |\vec{c}| &= \sqrt{(-1)^2 + 3^2 + 5^2} \\ &= \sqrt{1 + 9 + 25} = \sqrt{35} \end{aligned}$$

(e)

$$\vec{b} - \vec{a} = \langle -2, 5 \rangle - \langle 5, 3 \rangle = \langle -2 - 5, 5 - 3 \rangle = \langle -7, 2 \rangle$$

$$\begin{aligned} |\vec{b} - \vec{a}| &= \sqrt{(-7)^2 + 2^2} \\ &= \sqrt{49 + 4} = \sqrt{53} \end{aligned}$$

Exercise 2: Partial Differentiation

- Suppose that we have the following scalar field:

$$f(x, y, x) = xyz + yz + xz^2$$

Find all the partial derivatives of this function and write out the gradient vector i.e.

$$\nabla f = \left\langle \frac{\delta f}{\delta x}, \frac{\delta f}{\delta y}, \frac{\delta f}{\delta z} \right\rangle$$

- Remember that when calculating a derivative in say x, you assume other variables such as y or z remain constant.

Solution 2: Partial Differentiation

- We have the following scalar field:

$$f(x, y, x) = xyz + yz + xz^2$$

- First, we differentiate $f(x, y, z)$ with respect to x:
  - The derivative of xyz with respect to x is yz.
  - The derivative of yz with respect to x is 0 (since it does not contain x).
  - The derivative of xz^2 with respect to x is z^2

  Thus, we get:

$$\frac{\partial f}{\partial x} = yz + z^2$$

Solution 2: Partial Differentiation

- We have the following scalar field:

$$f(x, y, x) = xyz + yz + xz^2$$

- Second, we differentiate $f(x, y, z)$ with respect to y:
  - The derivative of xyz with respect to y is xz.
  - The derivative of yz with respect to y is z.
  - The derivative of xz^2 with respect to y is 0 (since it does not contain y).

  Thus, we get:

$$\frac{\partial f}{\partial y} = xz + z$$

Solution 2: Partial Differentiation

- We have the following scalar field:

$$f(x, y, x) = xyz + yz + xz^2$$

- Third, we differentiate $f(x, y, z)$ with respect to z:
  - The derivative of xyz with respect to z is xy.
  - The derivative of yz with respect to z is y.
  - The derivative of xz^2 with respect to z is $2xz$.

  Thus, we get:

$$\frac{\partial f}{\partial z} = xy + y + 2xz$$

Solution 2: Partial Differentiation

- Therefore, for the following scalar field:

$$f(x, y, x) = xyz + yz + xz^2$$

the gradient vector ∇f is

$$\begin{aligned}\nabla f &= \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle \\ &= \left\langle yz + z^2, \; xz + z, \; xy + y + 2xz \right\rangle\end{aligned}$$

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